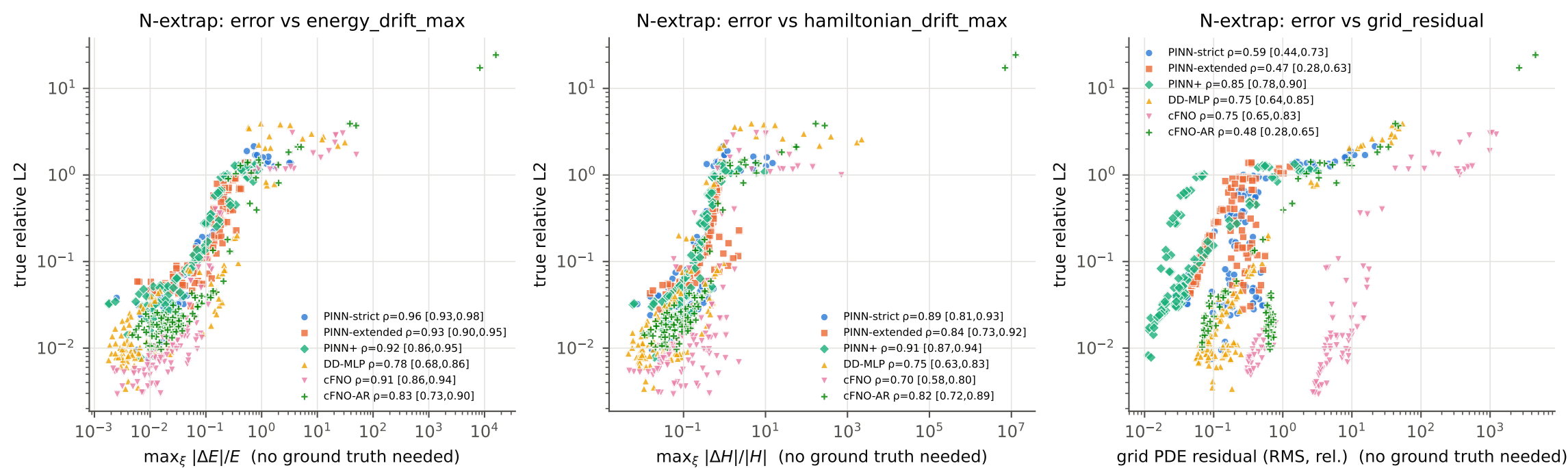

N-extrap: error vs energy_drift_max
true relative L2
max$_\xi$ |ΔE|/E (no ground truth needed)
PINN-strict ρ=0.96 [0.93,0.98]
PINN-extended ρ=0.93 [0.90,0.95]
PINN+ ρ=0.92 [0.86,0.95]
DD-MLP ρ=0.78 [0.68,0.86]
cFNO ρ=0.91 [0.86,0.94]
cFNO-AR ρ=0.83 [0.73,0.90]
N-extrap: error vs hamiltonian_drift_max
true relative L2
max$_\xi$ |ΔH|/|H| (no ground truth needed)
PINN-strict ρ=0.89 [0.81,0.93]
PINN-extended ρ=0.84 [0.73,0.92]
PINN+ ρ=0.91 [0.87,0.94]
DD-MLP ρ=0.75 [0.63,0.83]
cFNO ρ=0.70 [0.58,0.80]
cFNO-AR ρ=0.82 [0.72,0.89]
N-extrap: error vs grid_residual
true relative L2
grid PDE residual (RMS, rel.) (no ground truth needed)
PINN-strict ρ=0.59 [0.44,0.73]
PINN-extended ρ=0.47 [0.28,0.63]
PINN+ ρ=0.85 [0.78,0.90]
DD-MLP ρ=0.75 [0.64,0.85]
cFNO ρ=0.75 [0.65,0.83]
cFNO-AR ρ=0.48 [0.28,0.65]